Question-1

Suppose,

Rawalpindi=A, Attock=B, Chakwal=C, Jhelum=D, SArgodha=E

Mandi=F, Khushab=G, Gujranwala=H, DGK=I, Jhang=J

Faisalabad=K, Lahore=L, Multan=M, Sheikhupura=N, OKara=O,

Bahawalpur=P

$A \rightarrow M$ $\quad A^* \Rightarrow f(n) = g(n) + h(n)$

$g(n)$: actual cost, $h(n)$: estimated cost

Step	Frontier (Node, f(n))	Expand	Explored
0	(A, 350)	∅	∅
1	(A-B, 470), (A-C, 400), (A-D, 390)	A	{A}
2	(A-B, 470), (A-C, 400), (A-D-F, 460)	D	{A, D}
3	(AB, 470), (ADF, 460), (ACG, 660)	C	{A, D, C}
4	(AB, 470), (ACG, 660), (ADFH, 600)	F	{A, D, C, F}
5	(ACG, 660), (ADFH, 600), (ABE, 920)	B	{A, D, C, F, B}
6	(ACG, 660), (ABE, 920), (ADFHL, 840)	H	{A, D, C, F, B, H}
7	(ABE, 920), (ADFHL, 840), (ACGI, 860)	G	{A, D, C, F, B, H, G}
	(ACGP, 810), (ACGM, 670), (ACGJ, 860)		
8	(ABE, 920), (ADFHL, 840), (ACGI, 860)	M, GOAL	{A, D, C, F, B, H, G, M}
	(ACGP, 810), (ACGJ, 860)		

Final Path: A(Rawalpindi) → C(Chakwal) → G(Khushab) → M(Multan)

Cost : (250+290+130)+0 ⇒ 670

Question-2

a

No. It overestimates the actual cost to goal for at least 1 Node. Therefore, it is not admissible.

Ex.

- Jhang → Multan COST = 50
- Heuristic value of Jhang = 130

→ Overestimated heuristic is never admissible.

b

No. It is not consistent. ∵ $h(A) \leq c(A, C) + h(C)$

But this condition is violated in Jhang → Multan.

$h(\text{Jhang}) \leq c(\text{Jhang, Multan}) + h(\text{Multan})$

$130 \not\leq 50 + 0$ (False)

Hence, The hueristic violates the consistency condition and is not consistent.

c

If a heuristic overestimates the actual cost, A* may select a path that appears cheaper but is actually more expensive. It can lose its optimality guarantee, mean it may not find its shortest path.

Question-3

State representations

A state is represented as → (A,B) where

A = Amount of water in jug A

B = Amount of water in jug B.

→ Initial state ⇒ (0,0)

→ Goal state ⇒ Any state where A=2 or B=2

Operators

1. Fill jug A completly
2. Fill jug B completly
3. Empty jug A
4. Empty jug B
5. Pour water from A to B
6. Pour water from B to A

Step	Frontier	Expand	Explore
0	(0,0)	ϕ	ϕ
1	(4,0),(0,3)	(0,0)	(0,0)
2	(0,3),(4,3),(1,3)	(4,0)	(0,0),(4,0)
3	(4,3),(1,3),(3,0)	(0,3)	(0,0),(4,0),(0,3)
4	(1,3),(3,0)	(4,3)	(0,0),(4,0),(0,3),(4,3)
5	(3,0),(1,0)	(1,3)	(0,0),(4,0),(0,3),(4,3),(1,3)
6	(1,0),(3,3)	(3,0)	(0,0),(4,0),(0,3),(4,3),(1,3),(3,0)
7	(3,3),(0,1)	(1,0)	(0,0),(4,0),(0,3),(4,3),(1,3),(3,0),(1,0)
8	(0,1),(4,2)	(3,3)	(0,0),(4,0),(0,3),(4,3),(1,3),(3,0),(1,0),(3,3)
9	(4,2),(4,1)	(0,1)	(0,0),(4,0),(0,3),(4,3),(1,3),(3,0),(1,0),(3,3),(0,1)
10	(4,1)	(4,2)	Goal found

Final Path

- (0,0) → (0,3) → (3,0) → (3,3) → (4,2)
- Initial state → Fill jug B → Pour B into A → Fill jug B → Pour B into A

Question-4

DFS:

Explores one path as deeply as possible, before backtracking. It does not consider path cost.

UCS:

Always selects the path with lowest total cost, so it prefers the deep and cheap path.

Greedy:

Selects the Node that appears closest to the goal using the heuristic value **h(n)**. It ignores the actual path cost.

A*:

Considers both actual path cost and estimated remaining cost using **f(n) = g(n) + h(n)**. It finds the least cost path hence deep path.

Question-6

Start : Lahore

Goal : Islamabad

$$f(n) = g(n) + h(n)$$

($h(n)$ → estimated cost; $g(n)$ → actual cost)

Step	Frotier (Nodes, f(n)	Expand	Explored
0	(Lahore, 12)	ϕ	ϕ
1	(Lahore-Sahiwal, 10), (Lahore-Gujranwala, 13)	Lahore	Lahore
2	(Lahore-Gujranwala, 13), (Lahore-Sahiwal -Multan, 14)	Sahiwal	Lahore, Sahiwal
3	(Lahore-Sahiwal-Multan, 14), (Lahore-Gujranwala-Rawalpindi, 16)	Gujranwala	Lahore, Sahiwal, Gujranwala
4	(Lahore-Gujranwala-Rawalpindi, 16), (Lahore-Sahiwal-Multan-Islamabad, 14)	Multan	Lahore, Sahiwal Gujranwala, Multan
5	(Lahore-Gujranwala-Rawalpindi, 16)	Islamabad (Goal)	Lahore, Sahiwal, Gujranwala, Multan, ~~Islamabad~~ Islamabad

Final path:

Lahore → Sahiwal → Multan → Islamabad

Cost:

$5 + 6 + 3 \Rightarrow 14$

$g(\text{Islamabad}) = 14$

Question-5

Mistake

The student ignored the actual path cost **g(n)**. **A*** must consider both actual **g(n)** and estemated **h(n)** costs to find the best path.

A* formula

$$f(n) = g(n) + h(n)$$

What algo does it becomes?

It becomes Greedy best-first Search, because it selects nodes based on only the heuristic values **h(n)**.

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